

2-Adapter LoRA Merging Report

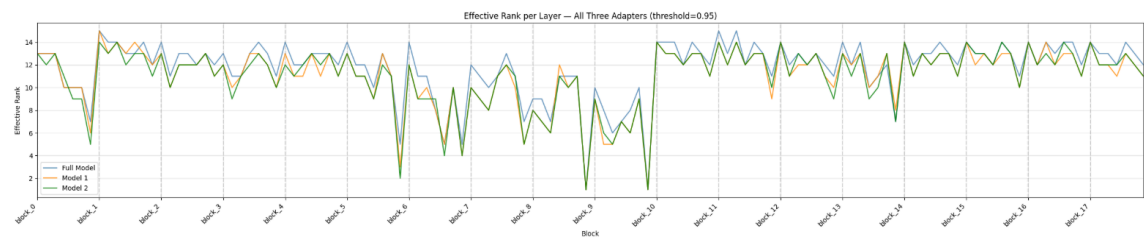
Experiment 1: Merging Adapters Trained on Disjoint Subsets of the Same Dataset

Experimental Setup

Parameter	Details
Dataset	AG-News (50,000 samples)
Full Model Training	Trained on all 50k samples
Adapter 1 Training	Trained on first 25k samples (disjoint subset)
Adapter 2 Training	Trained on second 25k samples (disjoint subset)
Base Model	Gemma-3-270M
Merging Goal	Combine two partial adapters to approximate full-data training

Effective Rank Analysis

The plot below shows the effective rank per transformer block for all three adapters (Full Model, Model 1, Model 2), computed at a 95% singular value threshold. Effective rank reflects how many singular value dimensions carry meaningful information in each layer's LoRA update matrix.



All three adapters maintain closely aligned effective rank profiles across blocks, generally fluctuating between 8 and 15. The tight tracking between Model 1 and Model 2 confirms that both subsets of data drive similar representational complexity per layer, which is a favorable sign for merging — the adapters are exploring the same subspaces and should combine without destructive interference.

Merging Methods — Explanation

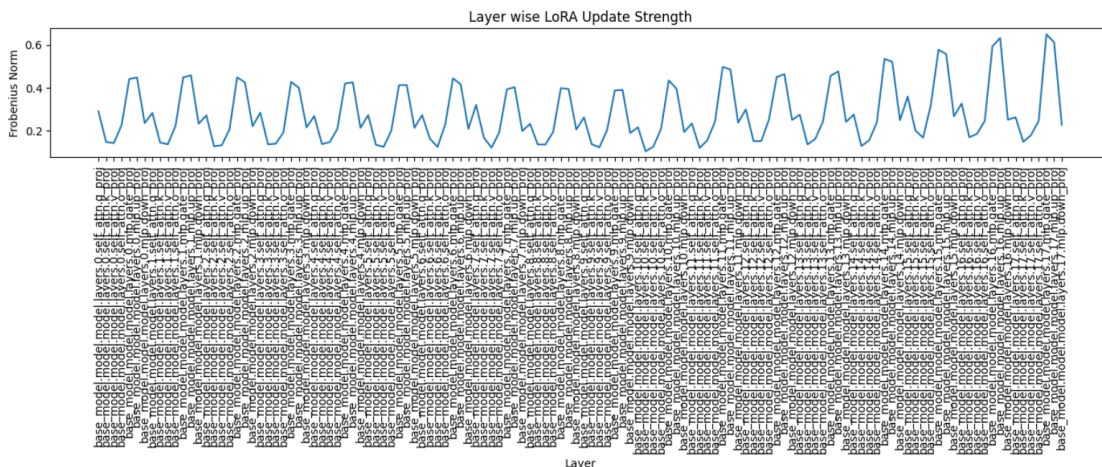
Five methods were applied to merge Adapter 1 and Adapter 2. The table below explains how each works and its suitability for this scenario.

Method	How It Works	Strengths / Weaknesses
Linear	Computes a weighted average of adapter weights. Each adapter contributes a fixed coefficient (e.g., 0.5 each).	Simple and fast. May dilute task-specific signals if adapters diverge significantly.
SVD	Applies Singular Value Decomposition to the merged delta matrix and truncates low-rank components to clean up noise.	Better at preserving dominant directions. Slight improvement over linear by removing noise.
TIES	Trims low-magnitude parameters, resolves sign conflicts by majority vote, then averages only the elected parameters.	More principled conflict resolution. Effective when adapters share task structure but differ in sign.
DARE	Randomly drops a fraction of delta parameters (density=0.2 means only 20% survive) then rescales the remainder.	Designed for large, dense deltas. Performs poorly here because 25k-sample adapter deltas are already sparse.
SLERP	Spherical Linear Interpolation. Merges adapters along the geodesic on the weight manifold rather than a straight average.	Best at preserving angular structure of weights. Closest to full-model performance in this experiment.

Frobenius Norm of Merged Models — Experiment 1

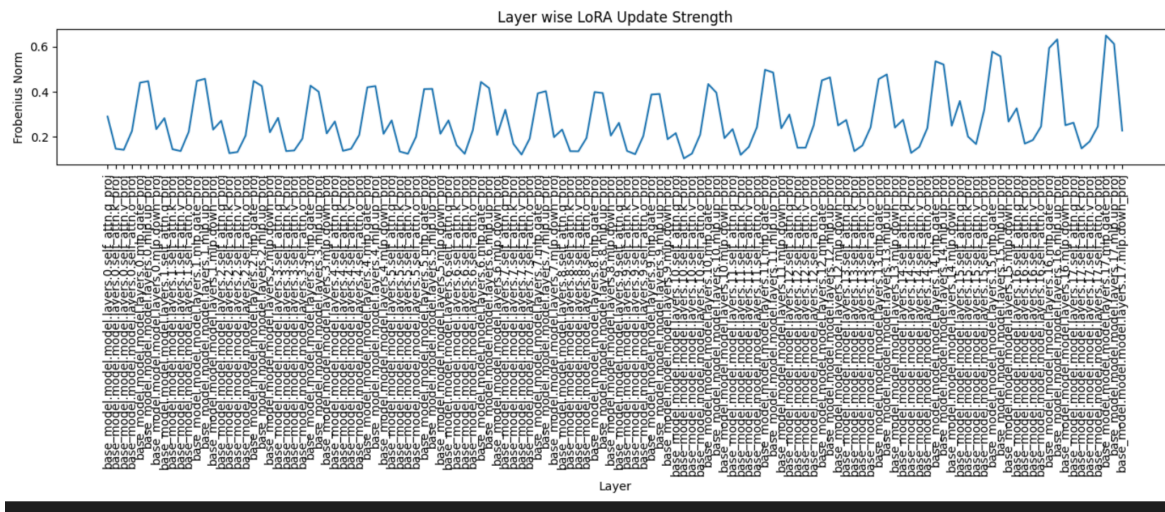
The Frobenius norm of the LoRA delta matrix per layer is a measure of update strength — how much a given layer was shifted from the base model weights during fine-tuning. Peaks in the plots correspond to layers that learned more task-specific information, while troughs indicate layers that changed minimally.

Linear Merge



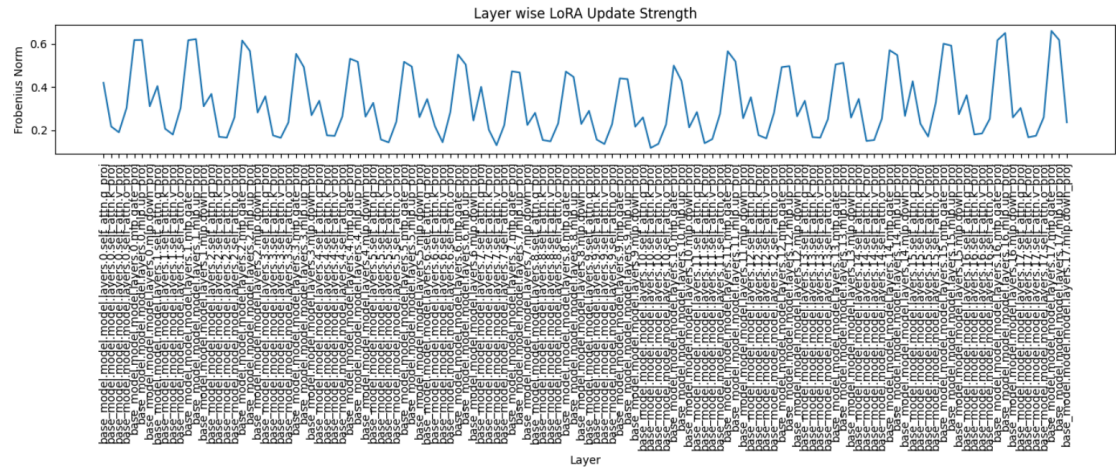
The linear merged model shows a stable oscillating pattern with norms in the 0.2–0.6 range. The rhythm alternates between attention and MLP sub-layers, with MLP peaks slightly higher. The smooth, consistent envelope across all 18 blocks indicates that linear averaging preserves the overall magnitude structure of both source adapters without amplifying any particular layer.

SVD Merge



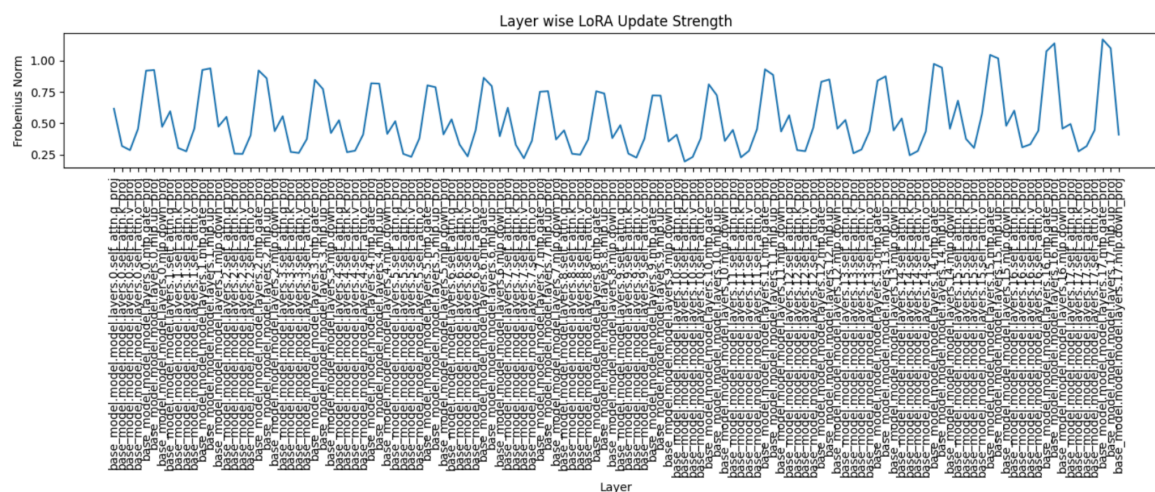
The SVD merged model displays a similar oscillating pattern to linear but with slightly elevated peaks in early layers (blocks 0–2) and more variation toward later layers. The low-rank truncation applied after averaging removes noise components, which can slightly inflate or deflate individual layer norms depending on how much energy sits in the discarded singular values. Overall, the profile remains well-behaved and close to the linear model.

TIES Merge



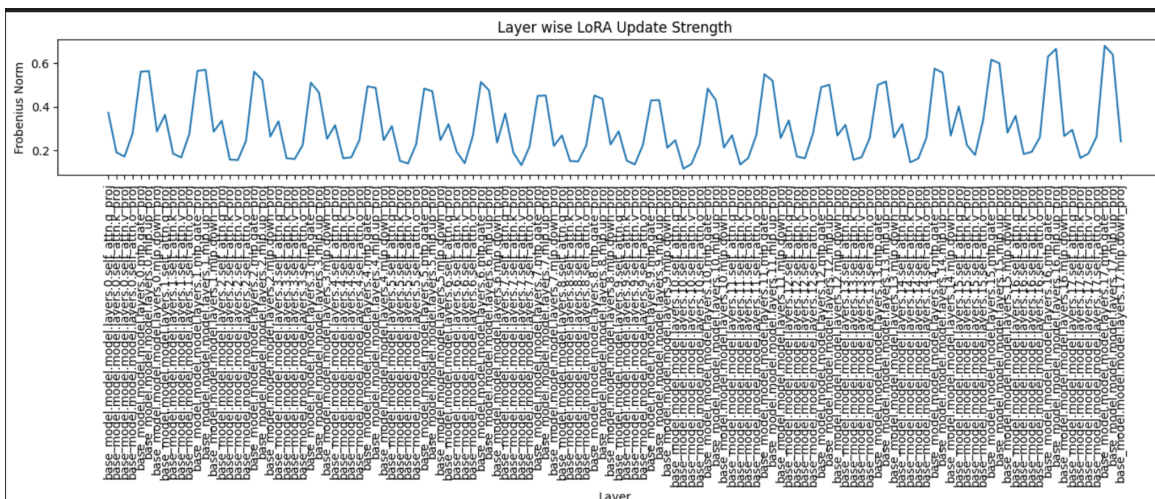
TIES produces a profile similar in shape to linear and SVD, with norms bounded in the same 0.2–0.6 range. Because TIES zeros out low-magnitude and conflicting parameters before averaging, some layers see slightly reduced norms where sign conflicts were common. The profile is still smooth, confirming that TIES does not introduce erratic layer-wise update magnitudes.

DARE Merge



The DARE merged model shows a dramatically different profile: norms range from 0.6 to over 1.5, roughly 2-3x higher than the other methods. This inflation is a direct consequence of DARE's rescaling step — after randomly dropping 80% of parameters (density=0.2), the surviving 20% are upscaled to preserve expected magnitude. With already-sparse 25k-sample adapter deltas, this rescaling overcorrects and produces unnaturally large weight updates, which explains DARE's poor accuracy in this experiment.

SLERP Merge



SLERP shows norms with a clear upward trend as depth increases — later layers accumulate stronger updates. This growing norm pattern is consistent with SLERP's geometric interpolation: by travelling along the manifold between two adapter points rather than averaging coordinate-by-coordinate, SLERP can produce merged weights whose magnitude exceeds either source adapter individually. The steady growth toward deeper layers suggests the merge is concentrating representational changes where they matter most.

Results

All merged models were evaluated on the AG-News test split using Accuracy and macro-averaged F1 Score. The full model (trained on all 50k samples) serves as the upper-bound baseline.

Model	Accuracy	F1 Score	Rank vs Full
gemma-3-270m-slerp-merged	0.8620	0.8510	1st (Best)
gemma-3-270m-ties-merged	0.8480	0.8393	2nd
gemma-3-270m-svd-merged	0.8440	0.8357	3rd
gemma-3-270m-linear-merged	0.8420	0.8332	4th
gemma-3-270m-dare-merged	0.7580	0.7365	5th (Worst)
full_model (reference)	0.8720	0.8690	Baseline

Key Observations

- SLERP achieves the closest approximation to the full model, trailing by only ~1 percentage point in accuracy (0.862 vs 0.872). Its geodesic interpolation preserves the angular structure of weight matrices better than flat averaging.
- TIES ranks second. Its sign-conflict resolution step removes interference between the two adapters, making it more principled than plain linear merging.
- SVD and Linear are competitive, with SVD slightly ahead due to its low-rank truncation cleaning up redundant components from the averaged delta matrix.
- DARE performs significantly worse (0.758 accuracy). With only 25k training samples per adapter, the weight deltas are already sparse — DARE's additional random dropout discards critical signal rather than noise, and its aggressive rescaling inflates norms to 2-3x normal levels.
- Attention q_proj and k_proj layers consistently show low cosine similarity between the merged model and the full model, suggesting that query and key projections are the hardest parameters to align across adapters. v_proj and MLP layers merge more faithfully.

Layer-wise Cosine Similarity: Linear Merged vs Full Model

The table below shows, for each transformer layer examined, how similar the merged (linear) adapter weights are to the full-model weights. Values closer to 1.0 indicate near-identical weight directions.

Layer	Cosine Similarity (Linear vs Full)
layers.15.mlp.down_proj	0.7831
layers.15.mlp.gate_proj	0.7479
layers.15.mlp.up_proj	0.7366
layers.15.self_attn.k_proj	0.5391
layers.15.self_attn.o_proj	0.7389
layers.15.self_attn.q_proj	0.5705
layers.15.self_attn.v_proj	0.6819
layers.16.mlp.down_proj	0.7308

layers.16.mlp.gate_proj	0.7884
layers.16.mlp.up_proj	0.8139
layers.16.self_attn.k_proj	0.6591
layers.16.self_attn.o_proj	0.7668
layers.16.self_attn.q_proj	0.5673
layers.16.self_attn.v_proj	0.8614
layers.17.mlp.down_proj	0.8190
layers.17.mlp.gate_proj	0.8047
layers.17.mlp.up_proj	0.8318
layers.17.self_attn.k_proj	0.5795
layers.17.self_attn.o_proj	0.7947
layers.17.self_attn.q_proj	0.5142
layers.17.self_attn.v_proj	0.8190

MLP layers (up_proj, down_proj, gate_proj) consistently score above 0.73, while attention q_proj and k_proj hover between 0.51 and 0.66. This indicates that the query and key subspaces diverge significantly between adapters, likely because each adapter has learned task-specific attention patterns from its 25k subset.

Experiment 2: Merging Adapters Trained on Different Tasks

Experimental Setup

This experiment : two adapters are trained on semantically different tasks. The goal is to produce a single model that retains mathematical reasoning ability (from OpenMathReasoning) while also incorporating general instruction-following ability (from Finetome).

Parameter	Details
Base Model	Qwen3-0.6B
Task 1 (Reasoning)	Adapter trained on 15k samples of OpenMathReasoning dataset (mathematical chain-of-thought reasoning)
Task 2 (Non-Reasoning)	Adapter trained on 15k samples of Finetome dataset (general instruction following)
Evaluation Task	Mathematical reasoning benchmark (Exact Match, ROUGE-1, ROUGE-2, ROUGE-L)
Merging Goal	Create a single model capable of both reasoning and instruction following

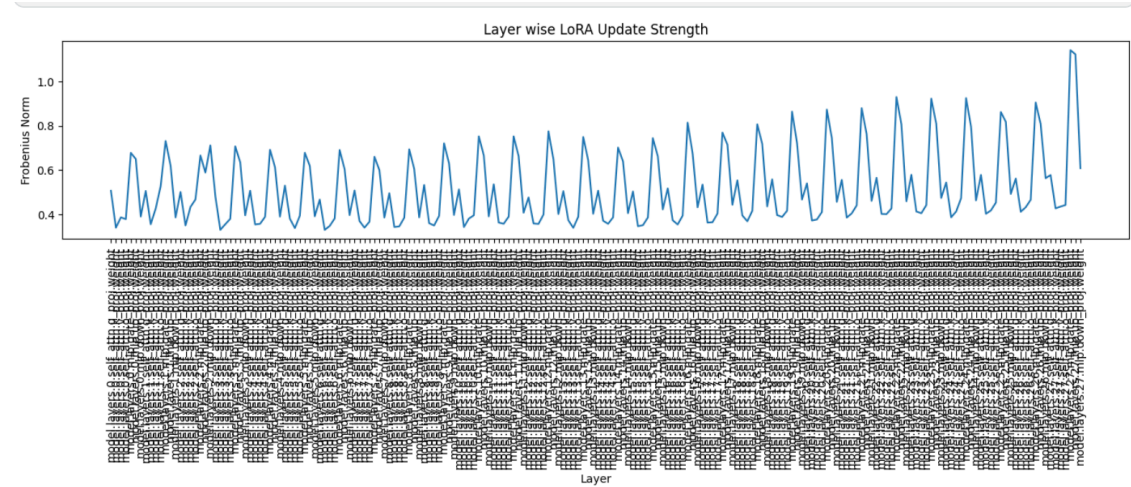
Evaluation Metrics

Metric	What It Measures
Exact	Exact Match — the fraction of predictions that exactly match the gold answer string. Strictest metric.
R-1	ROUGE-1 — overlap of individual words (unigrams) between prediction and reference.
R-2	ROUGE-2 — overlap of consecutive word pairs (bigrams). Captures phrase-level similarity.
R-L	ROUGE-L — longest common subsequence. Rewards output that preserves the reference word order.

Frobenius Norm of Merged Models — Experiment 2

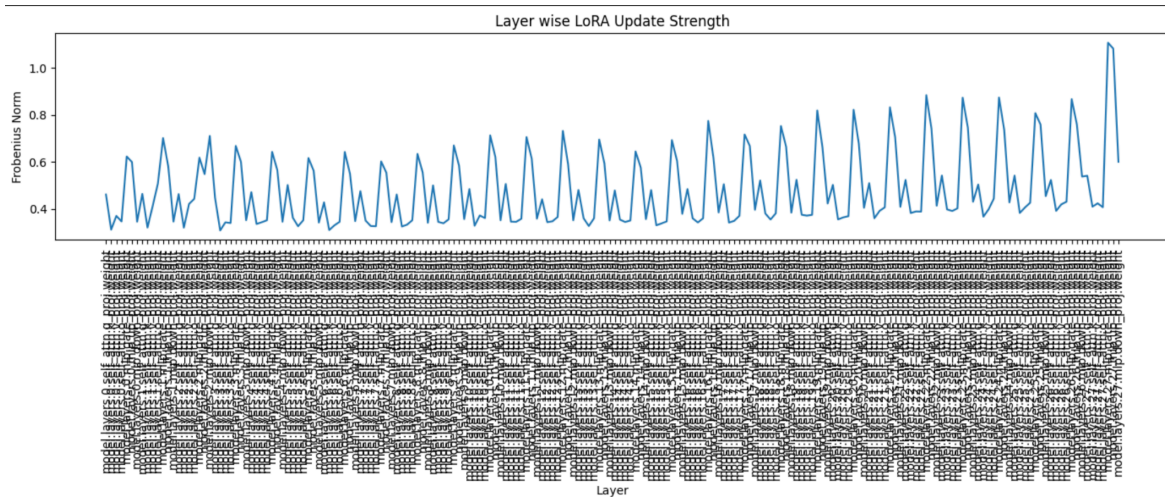
In this cross-task setting, the Frobenius norms are generally higher than in Experiment 1, reflecting the fact that adapters trained on semantically distant tasks tend to develop larger, more distinct weight deltas. The merging methods handle this increased delta magnitude differently.

Linear Merge



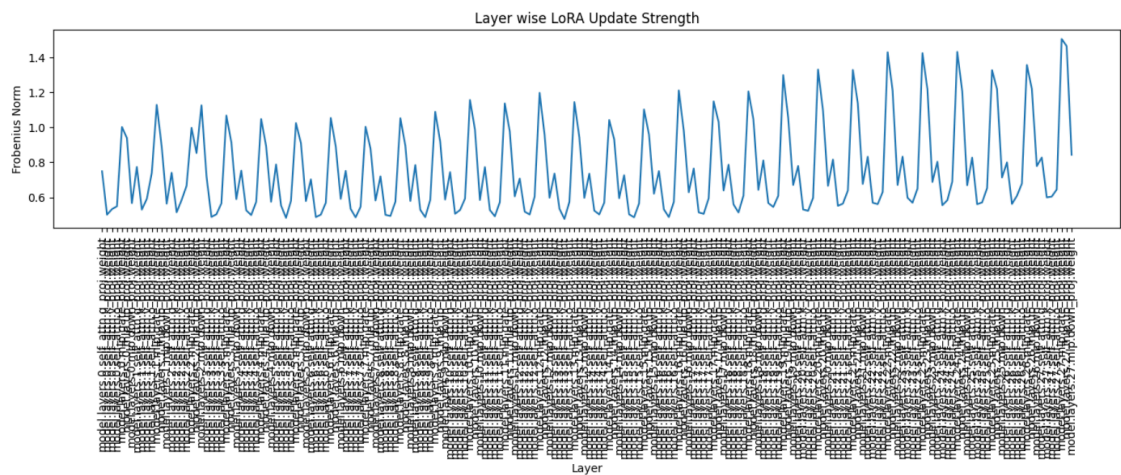
Linear averaging attenuates both adapters equally, which keeps norms modest but may wash out task-specific information from each adapter, particularly for reasoning-intensive layers.

SVD Merge



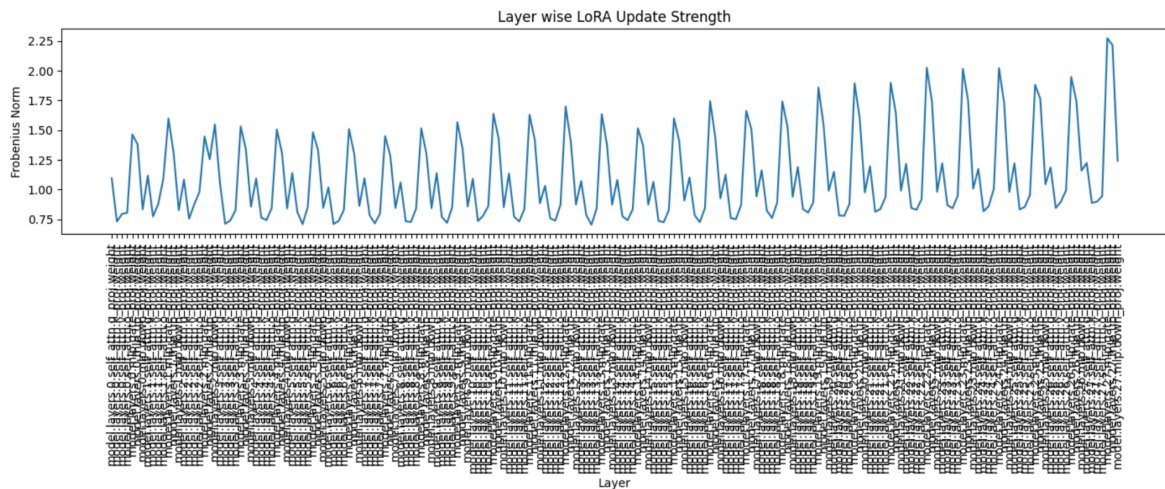
The SVD merge shows higher norms than linear, ranging from roughly 0.4 to 1.1, with norms growing toward later layers. This is explained by the fact that cross-task deltas are rich enough that most energy survives SVD's low-rank truncation — the top singular values carry genuine signal from both tasks, so little gets discarded.

TIES Merge



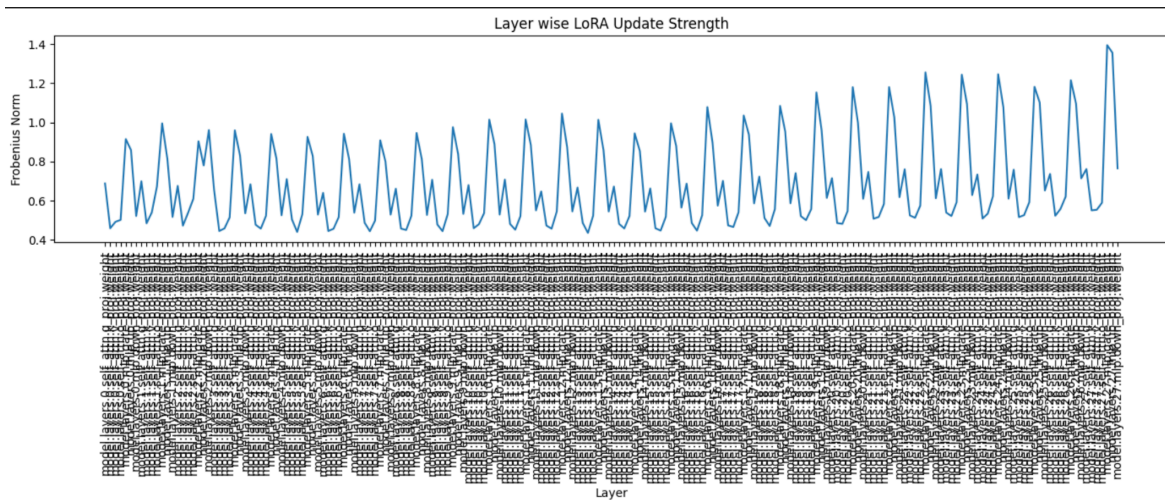
TIES produces the highest norms of any method in this experiment, ranging from 0.6 to 1.5. In the cross-task setting, the two adapters have extensive sign conflicts. TIES resolves these by zeroing out conflicting parameters and then rescaling survivors, which can paradoxically amplify the norm of the remaining parameters. The high norm and poor Exact Match score together suggest that TIES is selecting and amplifying the wrong subset of parameters for mathematical reasoning.

DARE Merge



DARE in the cross-task setting shows norms in the 0.75–2.25 range, growing consistently with depth — the largest profile of all methods. Unlike Experiment 1 where DARE's rescaling was destructive, here the cross-task adapter deltas are denser and larger, providing more meaningful signal for DARE to retain after its random drop step. The strong ROUGE scores from DARE in Experiment 2 support this: the model retains fluent generation even if exact answer precision is limited.

SLERP Merge



SLERP's cross-task Frobenius profile ranges from 0.4 to 1.4, with a clear increasing trend toward deeper layers. The geodesic path between the two task adapters in weight space produces merged parameters that grow in magnitude at depth, which is consistent with SLERP's known behavior of preserving the angular momentum of the source vectors. This structured growth likely contributes to SLERP's best-in-class Exact Match performance in Experiment 2.

Results

Results are compared against individual (non-merged) adapters as a reference.

Model	Exact	R-1	R-2	R-L
Qwen3-0.6b-slerp-merged	0.0850	0.3073	0.0703	0.1589
Qwen3-0.6b-dare-merged	0.0750	0.3226	0.0769	0.1650
Qwen3-0.6b-svd-merged	0.0700	0.3295	0.0781	0.1680
Qwen3-0.6b-linear-merged	0.0650	0.3291	0.0786	0.1665
Qwen3-0.6b-ties-merged	0.0400	0.3143	0.0727	0.1604
openmath (reference)	0.0700	0.3284	0.0762	0.1629
finetome (reference)	0.0500	0.3296	0.0780	0.1632

Key Observations

- SLERP achieves the highest Exact Match (0.085), actually exceeding the individual openmath adapter (0.070). This is a notable result: the merged model is better at exact answer prediction than the task-specific adapter alone.
- TIES performs worst on Exact Match (0.040). Its high Frobenius norms suggest it is amplifying the wrong parameters — sign-conflict resolution via majority vote may discard reasoning-critical weights that happen to conflict with instruction-following weights.
- Linear and SVD sit in the middle range, both outperforming the finetome adapter on Exact Match, showing that even simple merging preserves some mathematical reasoning.
- All merged models achieve ROUGE scores close to both individual reference adapters, indicating that fluency is largely preserved regardless of merging strategy. The main differentiator is Exact Match, which tests precision of reasoning outputs.

Comparative Summary Across Experiments

Dimension	Experiment 1 (Same Task, Split Data)	Experiment 2 (Different Tasks)
Best Merger	SLERP (Acc: 0.862)	SLERP (Exact: 0.085) / DARE (ROUGE)
Worst Merger	DARE (Acc: 0.758)	TIES (Exact: 0.040)
Gap to Baseline	SLERP loses only 1 percentage point vs full model	SLERP beats individual openmath model on Exact
Weak Layers	q_proj and k_proj show low cosine similarity vs full model	Cross-task merging amplifies attention head conflicts

Conclusions

- SLERP is the most consistently strong merging strategy across both experimental settings. Its geodesic interpolation preserves the geometric structure of weight spaces better than linear

approaches, and its Frobenius norm profiles show structured, depth-increasing update magnitudes rather than erratic behavior.

- TIES offers a good balance when adapters share task structure but differ in data. It suffers when tasks are semantically distant, as sign-conflict resolution discards critical cross-task parameters.
- `q_proj` and `k_proj` are the most fragile layers during merging. Future work should consider layer-selective merging strategies that apply different coefficients to attention subcomponents vs. MLP layers.
- Merging two adapters trained on half the data (Experiment 1) closely approximates full-data training performance with SLERP, suggesting that data-parallel adapter merging is a viable training strategy.